# Dual Data- and Knowledge-Driven Land Cover Mapping Framework for Monitoring Annual and Near-Real-Time Changes

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***Abstract*— As one of the most important application for remote sensing monitoring, land cover mapping has witnessed notable advancements in data acquisition, algorithmic diversity, and classification accuracy. Despite the instrumental role data-driven algorithms have played in the development of global land cover products, their inherent limitations as "black box" methods often fall short of meeting end-users' specific requirements. In this study, built upon the foundation of the earlier land cover monitoring platform [FROM-GLC plus(FGP)], a data and knowledge dual-driven framework (FGP 2.0) was developed as a user-adaptive framework for intelligent remote sensing land cover mapping. By incorporating ontology-based semantic descriptions with advanced data-driven algorithms, FGP 2.0 provides the capacity for both traditional annual mapping and emerging dynamic mapping. Our results illustrate that FGP 2.0 significantly improves the overall accuracy of annual maps by ∼5%, and dynamic maps by ∼20% compared to FGP. Moreover, an operational dynamic mapping tool has been developed on the Google Earth engine (GEE), enabling the generation of near-real-time land cover maps for any given place. With an extensible and flexible mapping framework, FGP 2.0 demonstrates the potential of customized land cover monitoring results to suit different application scenarios. This innovative approach not only meets the current demand for reliable annual and dynamic land cover maps but also sets a new benchmark for the integration of geoscientific expertise with machine learning techniques in remote sensing monitoring.**

***Index Terms*— FROM-GLC plus (FGP), knowledge-driven, machine learning, Sentinel-2.**

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## I. Introduction

ENHANCED by human activities and climate change, there has been a substantial and swift transformation in global land cover, which has significant impacts on global biodiversity, surface energy balance, and sustainable development [1], [2]. Accurate and reliable land cover datasets can provide an important database for earth system model development, resource management, and ecosystem evaluation [3], [4], [5]. Therefore, land cover mapping has always been one of the pivotal research directions in the field of remote sensing monitoring application.

The knowledge-driven method represented by the expert interpretation system is one of the mainstream remote sensing image classification and land cover mapping methods in the last century [6], [7]. Traditional knowledge-driven methods utilize the existing expert knowledge in the field of geosciences to achieve land cover classification by simulating the way of visual interpretation [8], [9]. The mapping results obtained can generally meet the monitoring demand and have a high degree of interpretability [10]. However, due

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to the limitation of remote sensing data acquisition and computational resources at that time, the imagery used to construct interpretation knowledge was relatively simple and lacked the multisource and time-series information [11], [12]. This kind of expert system was usually applicable to certain specific research areas and topics, and it was difficult to form a complete knowledge system that could be used in the task of large-scale land cover mapping.

In recent years, the richness and diversity of remote sensing data have been substantially increased with the advent of the big data era and the development of remote sensing technology [13], [14]. Meanwhile, the data processing capability of computer hardware and artificial intelligence technology in the field of computer vision have also been developed and made important breakthroughs [15], [16]. Therefore, data-driven remote sensing information intelligent extraction methods have been widely used in land cover remote sensing mapping research [2], [17], [18], which generally rely on a large number of land cover samples for feature extraction and classification model training [19], [20]. Based on this data-driven framework, land cover products covering different spatial resolutions and periods have been developed, including:1) the 5-km global land cover dataset (GLASS-GLC) from 1982 to 2015 [21]; 2) the 300-m European space agency climate change initiative (ESA-CCI) product from 1992 to 2020 [22]; 3) the 30-m annual GLC (AGLC) from 2000 to 2015 [23] and the GLC_FCS30 from 1985 to 2020 (five-year interval) [24]; 4) the 10- and 30-m finer resolution observation and monitoring of GLC (FROM-GLC) products [2], [18], the 10-m ESA WorldCover product for 2020 and 2021 [25], [26], and the near-real-time dynamic world (DW) products [27]; and 5) the multiple spatio-temporal resolution FROM-GLC plus (FGP) [28].

Compared with traditional knowledge-driven methods, land cover mapping using data-driven methods has less human intervention and a higher degree of automation, which promotes the intelligent development of land cover remote sensing mapping [29]. However, this kind of method relies on a large number of training samples and the ability of machine learning algorithms to learn the input features to achieve land cover mapping, lacking consideration and application of the uniqueness of geoscientific knowledge, and has faced obvious bottlenecks in mapping accuracy and practicability [30], [31]. This determined that the transparency and interpretability of the classification model are relatively low, mostly lacking the desired ability to generalize and resist interference [32], [33].

Faced with the limitations in land cover mapping research using data-driven methods, knowledge-driven algorithms have regained attention [34], [35]. Mapping approaches that integrate knowledge inference with data learning have emerged in recent years [36]. The most common and widely applied method currently is the integration of geographic object-based image analysis (GEOBIA) [37], [38], [39]. GlobeLand30 is one of the representative examples, combining pixel- and object-based classification with knowledge-based interactive verification to produce a global land cover mapping product [17]. While GEOBIA methods can mitigate unreasonable classification outcomes from data-driven algorithms, they fundamentally rely on the reclassification of image objects segmented from remote sensing data [40], [41], [42]. This reclassification, whether data-driven or manually determined, still carries inherent drawbacks and the massive workload of manual visual discernment. Moreover, object-based methods, while reducing the "salt-and-pepper" effect through segmentation, may result in the loss of details in the classification results [43], [44]. Therefore, inspired by the classification system from land cover classification system (LCCS) [45], a knowledge-driven mapping framework based on ontology has recently been introduced [46]. This framework, equipped with automatically generated semantic descriptions of Sentinel-2 spectral signature time series, offers an adaptable land cover mapping approach for end-users. At present, the fusion of ontology-based knowledge-driven and data-driven algorithms completes the active guidance and intelligent assistance of geoscientific knowledge mapping for interpreting remotely sensed imagery [36], [47]. However, such methods are confined to experimental applications on local cases, lacking the actual production of land cover mapping products. Consequently, how to synergize knowledge reasoning and data learning to achieve intelligent remote sensing mapping of land cover, meeting the current practical demand for land cover mapping products, remains a research direction laden with both potential and challenges [29].

In summary, with the increasing and diversifying remote sensing imagery and the progress in artificial intelligence technology, the current state of land cover remote sensing mapping has seen improvements in data multisourcing, algorithmic diversity, and classification accuracy. However, existing mapping methods are still predominantly driven by supervised or unsupervised data-driven algorithms, encountering bottlenecks in accuracy and practicality. In response to these limitations, there is a hope that knowledge-driven mapping approaches can offer intelligent guidance and assistance [30], [31]. This inspired us to develop a knowledge and data dual-driven land cover mapping framework (FGP 2.0) based on FGP [28], combining the strengths of LCCS classification scheme [45], ontology-based framework [46], along with other land cover products, to provide both traditional annual mapping and emerging dynamic mapping. Through comparisons with other global land cover products and the development of an operational dynamic mapping tool on GEE, our proposed framework demonstrates promptness, efficiency, and reliability for global land cover monitoring.

## II. Study Area and Data

### A. Study Area

We chose 12 5° × 5° grids as the study area for the dual-driven land cover mapping framework development (Fig. 1), encompassing a range of biomes such as tropical–subtropical forests, temperate–boreal forests and woodlands, shrublands and shrubby woodlands, deserts and semi-deserts, savannas and grasslands, intensive land use, etc. [48]. These grids were strategically chosen to represent diverse ecological zones, considering the distribution of land cover classes in existing global land cover mapping products, and taking into account

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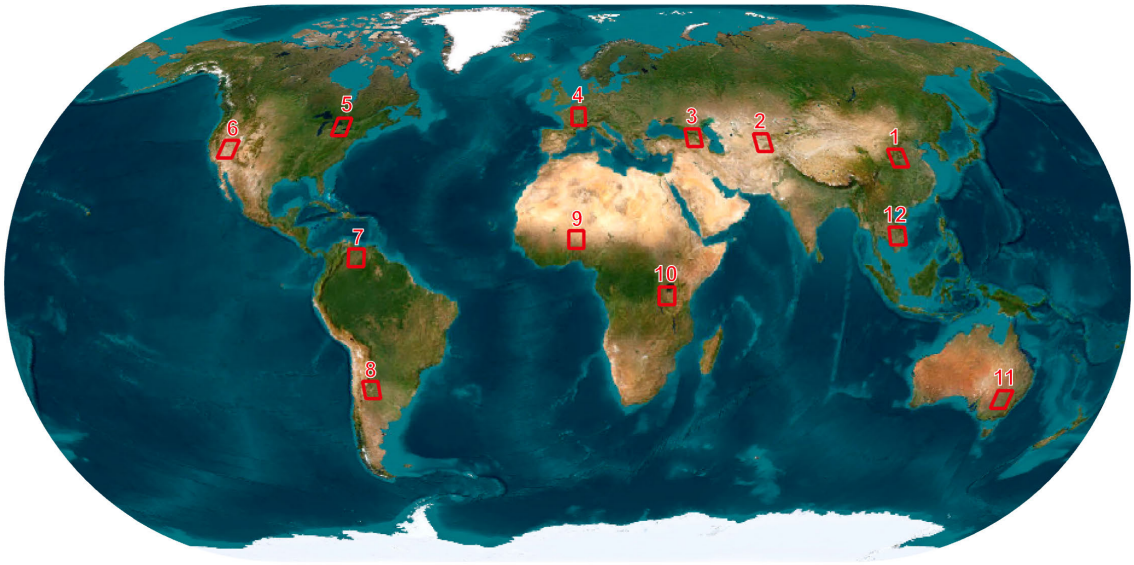



Fig. 1. Study area (12 5° × 5° grids): 1-East Asia; 2-Middle Asia; 3-East Europe; 4-West Europe; 5-North North America; 6-South North America; 7-North South America; 8-South South America; 9-North Africa; 10-South Africa; 11-Oceania; 12-Southeast Asia (Data source: Esri World Imagery).

our insights into the challenges of global land cover mapping research [28], [49]. The inclusion of these representative grids enhances the generalizability and applicability of the research results in large-scale land cover mapping, ensuring the robustness of the developed framework across various ecosystems and climatic conditions.

### B. Data

All Sentinel-2 top-of-atmosphere (TOA) imagery with less than 20% cloud cover in 2021 was used for FGP 2.0 annual mapping, while Sentinel-2 TOA imagery with less than 20% cloud cover in 2023 was used to validate the dynamic mapping capability of FGP 2.0. In addition, this study incorporates DW [27] and GlobeLand30 [17] products in the FGP 2.0 framework. The DW product, crafted through sample patches and a fully convolutional neural network (FCNN), ensures continuous and near-real-time land cover mapping in both spatial and temporal dimensions from June 27, 2015, to present, which allows it to be incorporated into the dynamic mapping framework of the FGP 2.0. For annual mapping, annual DW land cover maps can be derived employing a pixel-level annual mode strategy. As for the GlobeLand30 product, it was developed using a comprehensive approach that integrates pixel- and object-based methods with knowledge (POK-based) for the annual global land cover maps of 2000, 2010, and 2020. For this study, the annual updating of GlobeLand30 maps for 2021 was accomplished through an open and synergistic framework [50]. Therefore, both DW and GlobeLand30 products boast annual updating capabilities, providing indispensable information for the land cover mapping framework proposed in this study.

## III. Methods

The workflow for the proposed knowledge and data dual-driven land cover mapping method is illustrated in Fig. 2. Addressing both traditional annual mapping and the burgeoning dynamic mapping challenges, this framework provided the corresponding mapping strategy, utilizing data-driven FGP [28], knowledge-driven ontology-based framework [46] inspired by LCCS classification system [45], and existing land cover products. For the annual land cover mapping, the mapping results from FGP would be optimized by the classes of color-material and geographic minor entities combined with the DW [27] and GlobeLand30 [17] products specific to the corresponding year. For the dynamic land cover mapping, the classification outcomes from Sentinel-2 single scenes in FGP were refined using the updated FGP annual land cover map and classes of color-material, synergized with DW products. The evaluation of final land cover maps, compared with other mapping strategy and land cover products, underscore the versatility of this approach in accommodating knowledge and auxiliary datasets for improved land cover mapping accuracy and reliability.

### A. Data- or Knowledge-Driven Land Cover Mapping

*1) Data-Driven Classification With FGP:* FGP [28] was developed based on the earlier global land cover monitoring framework, FROM-GLC [2]. Incorporating methods of multiseason sample migration, multisource data fusion, and Random Forest, FGP provides a capacity for land cover mapping with flexibility in spatio-temporal resolutions. Through the data-driven FGP framework, it becomes possible to generate near-real-time multitemporal maps spanning from annual to daily resolutions and multiresolution maps ranging from 30 m to sub-meter precision. Therefore, the current two main land cover mapping directions, which are traditional annual mapping with historical collected imagery and emerging dynamic mapping with near-real-time imagery, can be both achieved with FGP.

For this study, annual Sentinel-2 surface reflectance imagery of 2021, obtained through Google Earth engine (GEE) [15],

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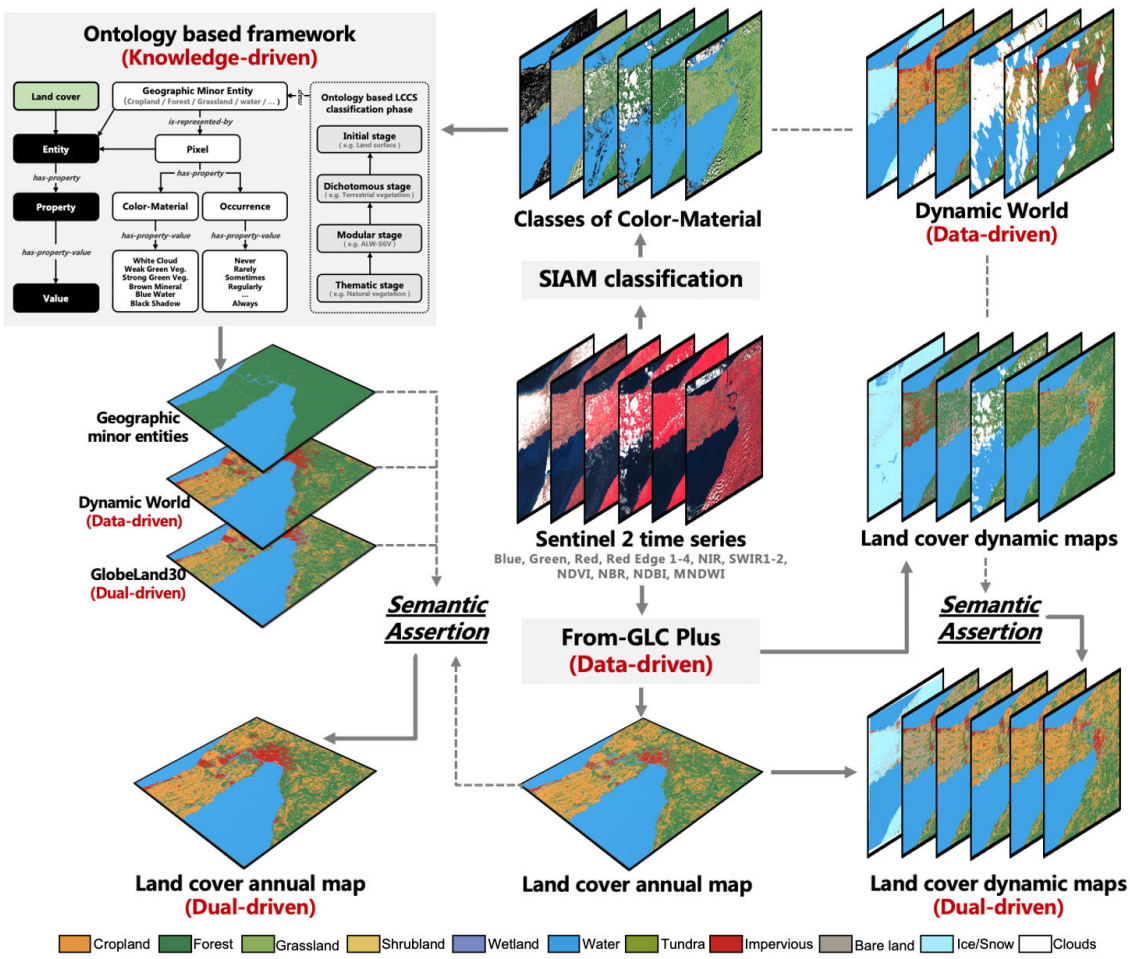



Fig. 2. Workflow of data and knowledge land cover mapping framework (FGP 2.0) developed in this study.

served as the foundation for FGP's annual land cover mapping. Additionally, continually updated Sentinel-2 surface reflectance imagery of 2023 was employed as the primary data source for near-real-time dynamic mapping. The spectral bands and indices, comprising blue, green, red, red edge 1–4, near infrared (NIR), shortwave infrared 1 (SWIR1), shortwave infrared 2 (SWIR2), normalized difference vegetation index (NDVI), normalized burn ratio (NBR), normalized difference built-up index (NDBI), and modified normalized difference water index (MNDWI), were utilized as features inputted into the Random Forest classifier for training. In the context of annual mapping, the feature matrix was developed by calculating the per-pixel 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, and 100 percentiles of each band. In the context of dynamic mapping, spectral and index features from individual Sentinel-2 scenes were extracted and utilized for the classification mapping of each scene. As a data-driven mapping framework, the land cover sample dataset plays a pivotal role in FGP. In this study, the migrated first all-season sample set (FAST) [51] and GLC estimation (GLanCE) samples [52] were employed to train the annual land cover classifier. Specifically, all GLanCE samples used had a high level of confidence (LC_Confidence$\geq$2) and were randomly split into two groups, with 70% allocated for training and the remaining 30% for validation. The daily migrated FAST samples were utilized for training the dynamic mapping classifier. Ultimately, the annual and dynamic land cover maps generated by FGP served as the foundational data-driven maps, forming the basis for the development of the knowledge and data dual-driven mapping framework in this research.

*2) Knowledge-Driven Classification With Ontology-Based Framework:* Despite the widespread application of data-driven classification algorithms in the field of land cover remote sensing mapping, they are still constrained by the limitations of "black box" methods that may not align with end-users' conceptualization of geographic features [32], [53], [54]. In light of this, a conceptual framework driven by knowledge was introduced to process Sentinel-2 time series, generating pre-classifications adaptable by end-users to meet their specific requirements [46]. This innovative approach was inspired by the LCCS classification system and scheme [45], and developed based on ontologies of scientific observation and geographic information, providing a descriptive foundation for representing geographic entities in remote sensing images.

For this study, an automatic knowledge-driven approach to classify Sentinel-2 imagery is developed with this ontology-based framework on GEE, following the application example provided by the original authors [46]. By bridging the gap

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between description of image characteristics and real-world characteristics in LCCS, this framework consists of the following classification stages. First, the spectral categories for each imagery in the Sentinel-2 annual time series were obtained using the satellite image automatic mapper (SIAM) spectral rules [55]. These SIAM spectral categories were then reclassified into six main classes of color-material, including "White Cloud," "Blue Water," "Weak Green Veg.," "Strong Green Veg.," "Brown Mineral," and "Black Shadow." Following this, per-pixel scores were calculated for each color-material class within the annual time series. These scores were derived from counts, with the cloud cover rate serving as the weighted factor for the corresponding image. The occurrences of each color-material class were then determined based on these scores, presenting the proportion (in %) of observations in the annual time series. To enhance clarity and maintain a natural language flow, these proportions were partitioned into eight categories, each labeled with adverbs of time to closely reflect natural language nuances: "Never," "Rarely," "Sometimes," "Regularly," "Predominantly," "Usually," "Often," and "Always." Subsequently, classification of geographic minor entities was processed. The initial and dichotomous stages were then realized based on the occurrences of each color-material to discriminate land surface material types: "Terrestrial Vegetation," "Terrestrial Non-Vegetation," "Aquatic Vegetation," "Aquatic Non-Vegetation," "Clouds," and "Shadows." The modular stage classes were finally obtained using the FGP 2.0, DW, and GlobeLand30 land cover maps based on the occurrences of color-material and dichotomous stage classes.

### B. Data and Knowledge Dual-Driven Land Cover Mapping

*1) Land Cover Annual Mapping:* Taking advantages of the combination of knowledge-driven ontology-based framework [46] and data-driven FGP [28], as well as the DW [27] and GlobeLand30 [17] land cover products, this study, firstly, developed a knowledge and data dual-driven classification framework for annual land cover mapping.

Specifically, the data-driven FGP annual land cover map for the year of 2021 was initially generated using the Random Forest classifier along with FAST [51] and GLanCE [52] samples (Fig. S1), following the methodology outlined in Section III-A1. Subsequently, scores and occurrences of color-material, along with the classification of geographic minor entities (dichotomous stage), were obtained using the framework introduced in Section III-A2. The modular stage classification of geographic minor entities was achieved and applied in conjunction with 2021 DW and GlobeLand30 land cover products to refine and optimize the FGP mapping results.

The combination of these two products for dual-driven annual land cover mapping was realized using the classes at dichotomous stage and the occurrence of color-material. Specially, the vegetation classes (cropland, forest, grassland, shrubland, tundra, and bare land) were optimized using the temporal occurrences of "Strong Green Veg.," "Weak Green Veg.," and "Brown Mineral," and double checked with DW and GlobeLand30. To achieve this goal, explicit assertions were established to describe these classes in symbolic natural language (Assertion 1), to optimize the FGP mapping results by setting the relation between the geographic entity of interest (e.g., bare land) and the pixel's description [46].

TABLE I
ACCURACY COMPARISON OF FGP 2.0 AND OTHER LAND COVER MAPPING RESULTS/PRODUCTS

| | Overall Accuracy (%) | Estimating Overall Accuracy (%) | Z-Score |
|---|---|---|---|
| FGP | 69.63 | 69.56 ± 2.79 | **3.69** |
| **FGP2.0** | **75.26** | **74.28 ± 2.66** | |
| FGP2.0[a] | 73.26 | 72.69 ± 2.68 | 1.38 |
| FGP2.0[b] | 62.18 | 67.95 ± 4.16 | **8.79** |
| FGP2.0[c] | 74.21 | 73.05 ± 2.68 | 0.69 |
| FGP2.0[d] | 71.73 | 71.70 ± 2.66 | **2.30** |
| FGP2.0[e] | 58.74 | 67.01 ± 5.01 | **10.71** |
| DW | 59.31 | 63.95 ± 3.15 | **9.84** |
| ESA WorldCover | 70.49 | 71.70 ± 2.74 | **2.95** |

[a]FGP+GlobeLand30+Knowledge; [b]FGP+DW+Knowledge; [c]FGP+GlobeLand30+DW; [d]FGP+GlobeLand30; [e]FGP+DW.

**Assertion 1.** *A ["Geographic Object"] initially classified as ["Geographic Minor Entity 1"] should be optimized as ["Geographic Minor Entity 2"], if that ["Geographic Object"] classified as ["Geographic Minor Entity 2"] in DW and/or GlobeLand30, and classified as [a value to qualify a color-material characteristic] or [class at dichotomous stage].*

Here, Assertions 2–4 were provided as examples:

**Assertion 2.** *A pixel initially classified as cropland should be optimized as bare land, if that pixel is classified as bare land in DW and GlobeLand30, and always classified as "Brown Mineral" color-material.*

**Assertion 3.** *A pixel initially classified as forest, grassland, or bare land should be optimized as shrubland, if that pixel is classified as shrubland in DW or GlobeLand30 and never classified as "Strong Green Veg.," and (at least) sometimes classified as "Weak Green Veg." color-material.*

**Assertion 4.** *A pixel initially classified as non-water should be optimized as water if that pixel is classified as water in DW or GlobeLand30, and classified as "Aquatic Non-Vegetation" at the dichotomous stage.*

In summary, the classes at the modular stage were determined based on adverbs of time derived from the temporal occurrences of color-material and classes at the dichotomous stage, in combination with information from DW and GlobeLand30, and further used to get the final land cover maps.

For the validation stage, the overall accuracy and $F1$-score for each land cover class were computed using the confusion matrix, employing migrated FAST validation samples [51] and the remaining 30% of GLanCE samples [52]. Additionally, overall accuracy with uncertainty was assessed through a calibration estimator based on an error matrix [56]. To quantify the significance of variations in mapping accuracy between our results and other products, Z-scores were computed (when $|Z - \text{score}| \geq 1.96$, at a 95% confidence interval, was considered indicative of statistical significance) [57].

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To assess the impact of incorporating knowledge, DW, and GlobeLand30 products on the final classification results, accuracy evaluations were conducted by comparing the mapping framework with knowledge + GlobeLand30 (FGP 2.0[a]), the framework with knowledge + DW (FGP 2.0[b]), the framework with GlobeLand30 + DW (FGP 2.0[c]), the framework with GlobeLand30 (FGP 2.0[d]), and the framework with DW (FGP 2.0[e]). The accuracy of DW was also quantified for comparative analysis. Additionally, considering the commendable performance and widespread recognition of WorldCover in global 10-m land cover mapping [26], [58], [59], [60], its accuracy for the year 2021 was also calculated as comparison. This validation approach thus provides a relatively comprehensive assessment of the effectiveness of the knowledge-driven framework and the influence of auxiliary land cover products.

*2) Land Cover Dynamic Mapping:* The all-season sample set FAST [51], featuring multitemporal labels and a sample migration method in FGP, has paved the way for generating near-real-time land cover maps based on continuously updated Sentinel-2 imagery. However, the intrinsic limitations of spectral features derived from single-scene remote sensing imagery, especially for land cover classes reliant on phenological characteristics reflected in annual spectral time series (such as cropland, forests, grasslands, etc.), make it challenging for the classification between these classes. In addressing this challenge within the previously proposed FGP, a time consistency check algorithm was employed to filter out unreasonable land cover changes [28]. Although effective, this method required significant computational resources for long time-series land cover maps and faced limitations in filtering and optimizing maps at the temporal endpoints.

Faced with these challenges, we introduced a novel dynamic mapping framework in FGP 2.0, integrating the updating annual land cover map produced by FGP and classes of color-material generated by the ontology-based framework. Specifically, for the single-scene image targeted for mapping, acquisition time and spatial coverage were, firstly, obtained. Based on the acquisition time as a reference, all Sentinel-2 images within the coverage for one year were traced back and then employed in FGP for the updating annual mapping. Subsequently, the ontology-based framework computes SIAM categories and color-material for the single-scene image. During the integration process, pixels categorized as "Weak Green Veg." and "Strong Green Veg." in the color-material class, corresponding to vegetation types (cropland, forest, grassland, shrubland, and tundra), are assigned the vegetation types from the updating annual land cover map. Symbolic natural language assertions similar with annual mapping are also built (e.g., Assertion 5–7).

**Assertion 5.** *A pixel initially classified as grassland should be optimized as cropland if that pixel is classified as cropland in updating annual map, and classified as "Strong Green Vegetation" or "Weak Green Vegetation" in color-material.*

**Assertion 6.** *A pixel initially classified as non-water should be optimized as water if that pixel is classified as water in updating annual map or DW, and classified as "Blue Water" in color-material.*

**Assertion 7.** *A pixel initially classified as non-impervious surface should be optimized as impervious surface if that pixel is classified as impervious in updating annual map or DW, and classified as "Brown Mineral" or "Black Shadow" in color-material.*

Aligning with the dynamic nature of the updating annual land cover map, DW dataset, and classes of color-material, this approach ensures that dynamic mapping results are not only reasonably optimized and corrected but also retain their inherent dynamic mapping characteristics.

## IV. Results

### A. Land Cover Annual Mapping Results

The comparison of classification accuracy, utilizing 1047 validation samples comprising FAST (608 points) and GLanCE (439 points) across all 12 study area grids (Fig. S2), is detailed in Table I. The integration of the knowledge-driven ontology-based framework with other land cover products enhances FGP mapping results, leading to a ∼5% improvement in overall accuracy. Moreover, the combination of both GlobeLand30 and DW yields higher mapping accuracy compared to using either data source alone (improved by ∼2% and ∼13%, respectively). Additionally, merging the geographic minor entities derived from knowledge-driven results for land cover product fusion outperforms accuracy solely reliant on existing products (improved by ∼1%). When compared with other land cover products, FGP 2.0 achieves ∼16% and ∼5% higher mapping accuracies than DW and WorldCover, respectively. Furthermore, Z-scores shown in Table I affirm the statistical significance of FGP 2.0's higher classification accuracy over current land cover products FGP, DW, and WorldCover at a 95% confidence level (Z-scores of 3.69, 9.84, and 2.95). This is further corroborated by Fig. 3, illustrating the superior mapping precision and stability of FGP 2.0. By computing the overall accuracy for the selected 12 grids, Fig. 3 presents the accuracy distribution through box plots. While the current FGP 2.0 classification strategy's accuracy is not significantly higher than FGP 2.0[a] and FGP 2.0[c] (with Z-scores of 1.38 and 0.69), its classification accuracy exhibits a more robust performance across all study areas. In addition, the comparison between FGP 2.0[a] and FGP 2.0[d], as well as between FGP 2.0[b] and FGP 2.0[e], also demonstrates the effectiveness of incorporating knowledge in improving classification accuracy and reducing uncertainty. This indicates that, by incorporating knowledge, geographic entities can better integrate existing land cover classification products, thereby improving the final classification accuracy and stability for global land cover classification applications.

To assess the classification accuracy across different land cover classes, $F1$-scores for each class were computed (Table II). FGP 2.0 exhibits a higher average $F1$-score than WorldCover data, with distinct improvements in certain land cover categories. The most pronounced accuracy enhancement is observed in shrubland classification, where FGP 2.0's $F1$-score outperforms FGP by 0.12. Furthermore, other

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TABLE II
*F*1-SCORE COMPARISON OF FGP 2.0 AND OTHER LAND COVER MAPPING RESULTS/PRODUCTS

| | FGP | FGP 2.0 | FGP 2.0[a] | FGP 2.0[b] | FGP 2.0[c] | FGP 2.0[d] | FGP 2.0[e] | DW | WorldCover |
|---|---|---|---|---|---|---|---|---|---|
| Cropland | 0.76 | **0.84** | 0.83 | 0.73 | 0.83 | 0.82 | 0.69 | 0.69 | 0.78 |
| Forest | 0.81 | **0.83** | 0.81 | 0.80 | 0.82 | 0.79 | 0.83 | 0.80 | 0.80 |
| Grassland | 0.59 | 0.61 | 0.59 | 0.40 | 0.60 | 0.60 | 0.34 | 0.29 | **0.62** |
| Shrubland | 0.50 | **0.62** | 0.57 | 0.46 | 0.61 | 0.58 | 0.43 | 0.36 | 0.42 |
| Wetland | 0.00 | 0.33 | 0.33 | 0.00 | 0.00 | 0.00 | 0.00 | 0.22 | **0.43** |
| Water | 0.83 | 0.85 | 0.82 | 0.85 | 0.77 | 0.78 | 0.78 | 0.78 | **0.94** |
| Impervious surface | 0.82 | **0.86** | 0.86 | 0.82 | 0.82 | 0.78 | 0.64 | 0.64 | 0.80 |
| Bare land | 0.64 | 0.62 | 0.59 | 0.09 | 0.59 | 0.51 | 0.06 | 0.55 | **0.68** |
| Average | 0.62 | **0.69** | 0.68 | 0.52 | 0.63 | 0.61 | 0.47 | 0.54 | 0.68 |

[a]FGP+GlobeLand30+Knowledge; [b]FGP+DW+Knowledge; [c]FGP+GlobeLand30+DW; [d]FGP+GlobeLand30; [e]FGP+DW.

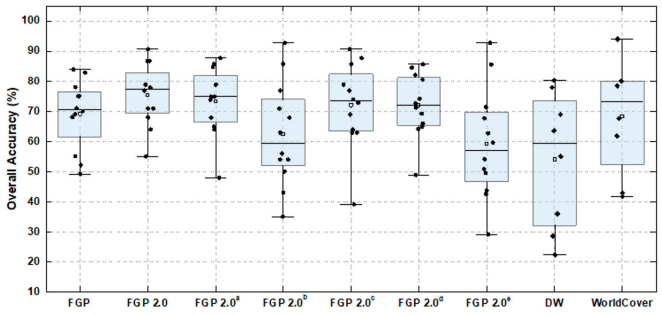



Fig. 3. Overall accuracy comparison with other land cover products for the selected 12 grids ([a]FGP+GlobeLand30+Knowledge; [b]FGP+DW+Knowledge; [c]FGP+GlobeLand30+DW; [d]FGP+GlobeLand30; [e]FGP+DW).

land cover classes also see improvements, such as cropland by 0.08 and impervious surface by 0.04. The land cover mapping comparison across the selected 12 grids further illustrates this point (Fig. 4). For instance, in South North America, the confusion between shrubland and bare land is significantly reduced using scores and occurrences of color-material ("Weak Green Veg.," "Strong Green Veg.," and "Brown Mineral,") alongside the classification of geographic minor entities ("Terrestrial Vegetation" and "Mineral"). Regions prone to misclassification between cropland and grassland, similar to North South America and Oceania, benefit from more accurate spatial distribution aided by visual interpretation of GlobeLand30 dataset. High-latitude areas, susceptible to snow/ice confusion during winter, also exhibit improvement, as seen in North North America and East Europe.

Fig. 5 illustrates the mapping details with zoomed-in view of the study area. We can find in Fig. 5(a) that, compared to FGP, FGP 2.0 provides more accurate water extent. With the assistant of ontology-based framework, the bounds of water are more precise than GlobeLand30 and DW. Additionally, while FGP exhibits fine-grained classification results for impervious surfaces, FGP 2.0 retains this advantage during the integration of DW and GlobeLand30 data by leveraging classes of color-material, resulting in more precise impervious surface classification. In regions with complex vegetation distribution, FGP 2.0 corrects misclassifications observed in FGP, such as agricultural land erroneously labeled as grassland [Fig. 5(b) and (c)]. The forest distribution between fields was retained in FGP 2.0 at the same time

TABLE III
ACCURACY COMPARISON OF FGP 2.0 AND OTHER LAND COVER DYNAMIC MAPPING RESULTS/PRODUCTS

| | Overall Accuracy (%) | Estimating Overall Accuracy (%) | Z-Score |
|---|---|---|---|
| FGP | 45.77 | 44.80 ± 2.49 | **12.28** |
| DW | 55.26 | 60.31 ± 2.82 | **5.63** |
| FGP 2.0 | **65.40** | **68.68 ± 2.85** | |

[Fig. 5(c)]. Correspondingly, forests that were misclassified as cropland in FGP were also corrected in FGP 2.0 [Fig. 5(d)]. This underscores that FGP 2.0, synthesizing the strengths of GlobeLand30 and DW while integrating knowledge-driven ontology-based frameworks, delivers more accurate and rational land cover mapping results.

### *B. Land Cover Dynamic Mapping Results*

Elaborating on the capabilities of the FGP 2.0 framework in dynamic land cover mapping, 12 scenes of 2023 Sentinel-2 imagery located in the 12 grids of study area were chosen to execute the dynamic classification procedure. With 1370 valid sample points interpreted visually, the dynamic mapping accuracy of the 12 scenes was calculated for FGP, FGP 2.0, and DW (Table III). Specifically, the dynamic mapping approach employed here with FGP did not undergo temporal consistency checks; instead, it directly generated migration samples and Random Forest classification results. According to the accuracy assessment results, FGP 2.0 reached significant higher accuracy compared with FGP and DW with Z-scores of 12.28 and 5.63, respectively. Through the combination of knowledge derived from the annual land cover map and ontology-based framework, the dynamic mapping accuracy of FGP witnessed a remarkable improvement of ∼20% in FGP 2.0. Additionally, FGP 2.0 demonstrates an enhanced accuracy of ∼10% compared to DW.

Furthermore, the outcomes of the classification process exhibited a noticeable improvement as illustrated in Fig. 6. Leveraging the annual land cover map as a foundation for vegetation distribution, the FGP 2.0 dynamic maps present more realistic depictions of vegetation dynamics, exemplified in regions such as East Europe, North South America, Oceania, and Southeast Asia. The annual map also contributes to rectifying the inaccurate classification of impervious surfaces, a result of the limited spectral information inherent in single-scene imagery. This discrepancy is evident in instances across

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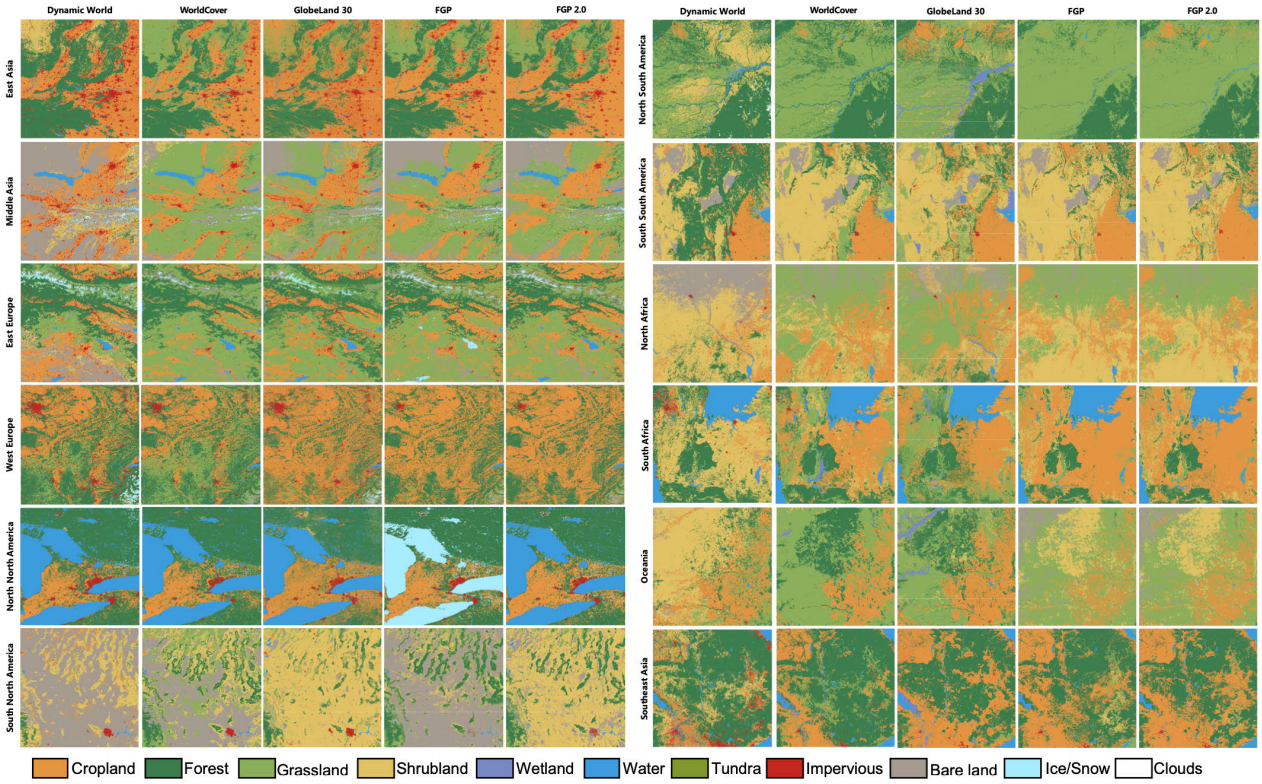



Fig. 4. Land cover mapping result comparison with other products for the selected 12 grids.

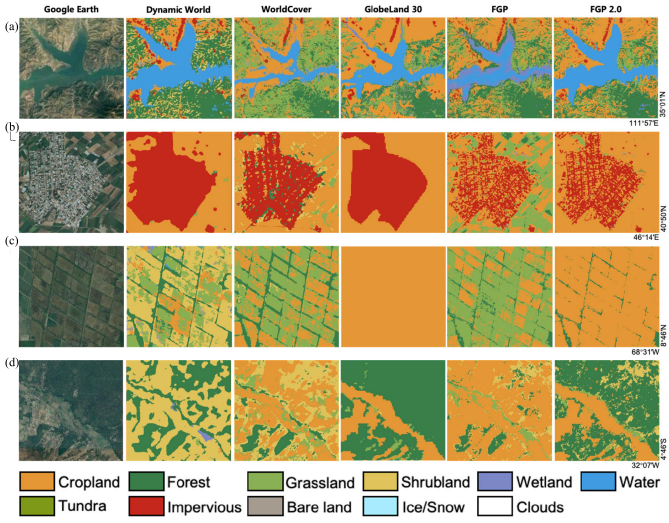



Fig. 5. Land cover mapping detail comparison with other products.

Middle Asia, South North America, and South South America. Notably, the misclassification of impervious surfaces has been ameliorated, as demonstrated in examples from West Europe and Oceania.

Additionally, the integration of results with classes of color-material plays a pivotal role in refining the overall classification outcomes. This is most evident in the substantial enhancement of water body classification, noticeable in examples from East Asia, North Africa, and South Africa. The FGP 2.0 framework also optimizes the representation of land cover in winter snow-covered regions by combining color-material, as depicted in examples from North North America. Moreover, this dynamic mapping framework effectively captures the seasonal variations in cropland, showcasing the periodic changes in non-crop-covered cropland, as evidenced in examples from West Europe. Furthermore, the 12 comprehensive land cover maps presented in Fig. 6 precisely span the 12 months of the year 2023. Consequently, the results underscore the FGP 2.0 framework's reliability in generating timely global land cover maps using continuously updated Sentinel-2 imagery.

To elaborate further on the dynamic mapping capabilities of the FGP 2.0 framework, we extended our analysis beyond the 12 selected grids to three additional regions, showcasing land cover changes throughout 2023 (Fig. 7). These changes were then compared with DW dataset, including natural seasonal shifts and human-induced alterations, such as ice melts—cropland recovery [Fig. 7(a)], seasonal variations in lakes [Fig. 7(b)], and the construction of large buildings [Fig. 7(c)]. When assessing single-scene mapping results, FGP 2.0 demonstrated a finer granularity in land cover classification compared to DW. It captured more precise and accurate details of buildings, roads, cultivated land parcels, and vegetation distribution. In terms of dynamic mapping over the course of the year, FGP 2.0 delivered more accurate representations of land cover changes. It effectively reflected the cultivation status of cropland, seasonal changes in water bodies, and the construction processes of emerging large buildings. These results highlight the framework's ability to provide reliable insights into monitoring land cover dynamics.

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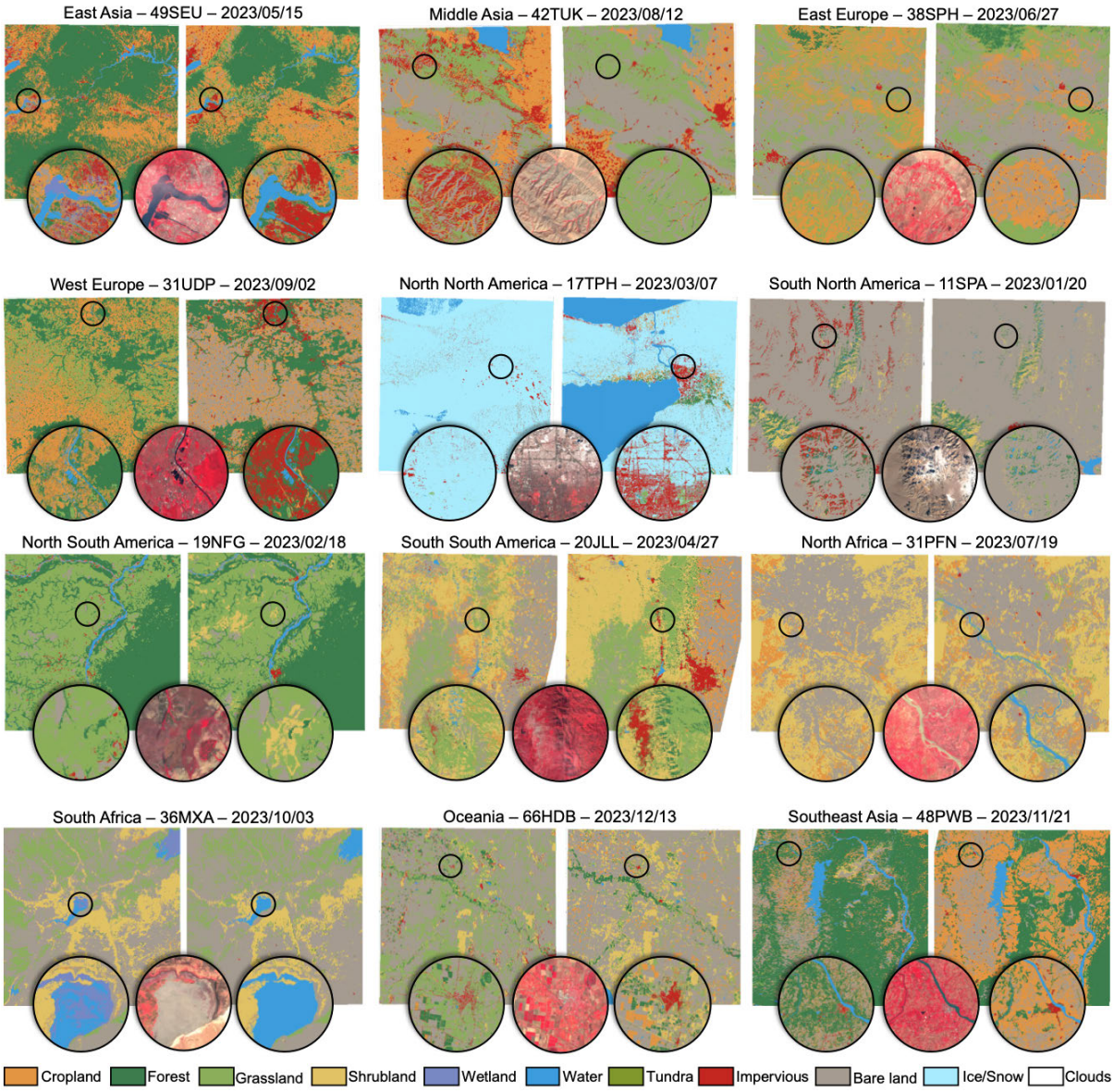



Fig. 6. Land cover mapping results of single Sentinel-2 image tiles. The left and right images are classification results obtained from FGP and FGP 2.0, respectively. The three circular images at the bottom represent the zoomed-in view classification results and the corresponding Sentinel-2 imagery (false-color rendering).

We further developed a near-real-time land cover mapping tool on GEE based on the FGP 2.0 land cover dynamic mapping framework (Fig. 8). This tool serves as a platform with on-the-fly computing capabilities, utilizing the latest Sentinel-2 imagery filtered by the map center to generate prompt and updated land cover maps. It can be used to explore near-real-time land cover information for any area of interest by zooming in/out on the map or using location searches. The tool also provides the comparison of local Sentinel-2 imagery (false-color rendering) and classification results through a convenient scrolling feature.

## V. Discussion

The knowledge and data dual-driven land cover mapping framework, FGP 2.0, proposed in this study was developed based on FGP [28], ontology-based framework [46] inspired by LCCS classification scheme [45], and existing annual land cover products [17], [27]. This framework was designed as a user-adaptive and extensible platform, aiming to provide traditional annual maps and the burgeoning dynamic maps. Integrating both knowledge-driven and data-driven classification procedures, the flexible FGP 2.0 can be extended beyond the optimization of FGP land cover products. To provide an example, here we deployed FGP 2.0 into the DW product pipelines.

Incorporating the optimization procedure into DW dataset involved first acquiring the Sentinel-2 imagery corresponding to each scene's classification results. Employing the ontology-based framework approach, the classes of color-material were then derived. The updating annual land cover maps for DW were subsequently generated using a per-pixel mode strategy based on the image acquisition time. Similar to

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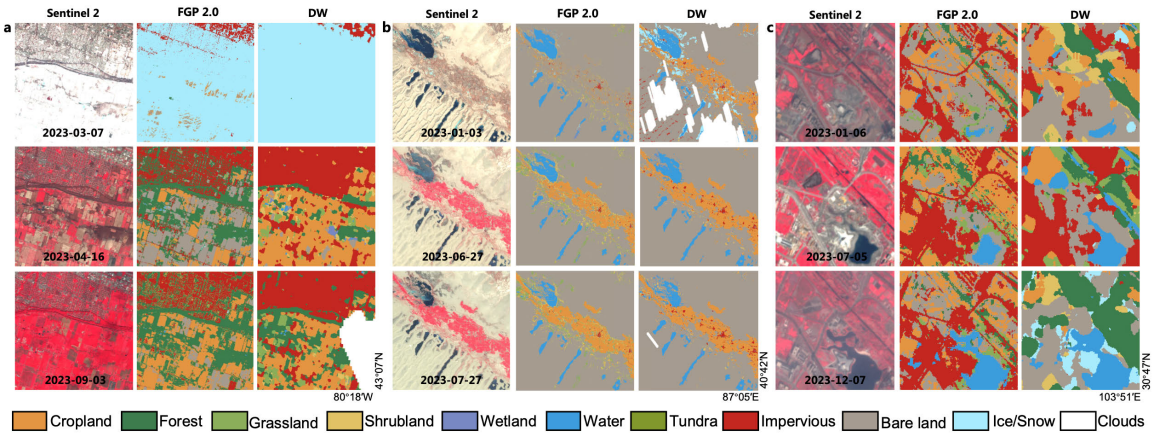



Fig. 7. Examples of near-real-time land cover dynamic mapping, including ice melts and cropland recovers, seasonal changes of lakes, new construction of large buildings (Sentinel-2 images are false-color rendered).

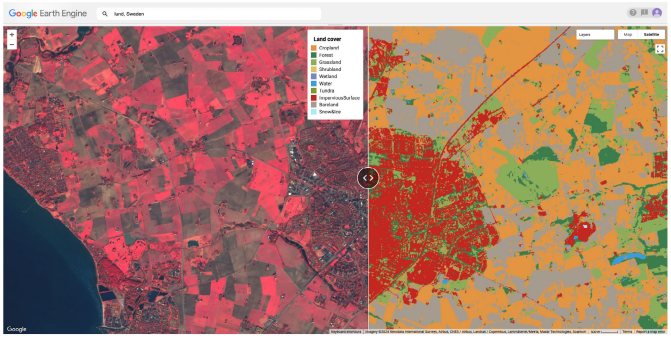



Fig. 8. Near-real-time land cover mapping tool on GEE based on the FGP 2.0 land cover dynamic mapping framework. The left map is false-color rendering Sentinel-2 image (captured on 31 Jan 2024 in Lund, Sweden), and the right map is the dynamic land cover mapping results for this image.

the process applied to FGP dataset, the optimized dynamic land cover maps (DW-FGP) were derived from the updating annual land cover map and color-material classes. As depicted in Fig. 9, this optimization resulted in refined details of building edges and vegetation distribution, and also provided the cultivation status of cropland throughout the year. Moreover, the optimization addressed instances of irrational land cover changes arising from insufficient information in single-scene imagery, including the correction of water body misclassifications due to mountain shadow, and the mitigation of building misclassifications, as illustrated in Fig. 9.

Moreover, with the user-adaptive and extensible mapping framework, FGP 2.0 not only enhances classification accuracy but also offers flexible solutions for diverse classification objectives. Considering the wide applications for land cover products across fields such as climate modeling, ecological monitoring, and socioeconomic analysis, the focus of land cover mapping has shifted from accuracy toward heightened practicality. Leveraging FGP 2.0, users can tailor classification results to regional monitoring tasks and different application scenarios by combining various existing land cover products and establishing distinct semantic assertions. Additionally, the classification system for dynamic land cover mapping and the definition of land cover changes can be adjusted according to various applications. By modifying the classification system and definitions, FGP 2.0 can be further used for monitoring seasonal vegetation changes and capturing sudden land cover changes caused by wildfires or extreme weather events. FGP 2.0, thus, showcases the potential to integrate diverse class definitions to synergize the increasingly rich landscape of land cover products, thereby supporting the multifaceted applications of land cover maps.

Despite its effectiveness in annual and dynamic land cover mapping, FGP 2.0 still has some limitations that need to be improved in the future. Firstly, the integration of existing land cover products can assist the ontology-based framework in interpreting land cover at the modular stage during the annual mapping task, optimizing results derived solely from machine learning algorithms. However, this integration may introduce errors from existing land cover products, and differences in classification systems among various datasets may lead to misjudgments in the fusion process. If the accuracy of the fused land cover product itself is not high, it could potentially compromise the precision of the original land cover classification results. This explains why FGP $2.0^{a}$ (FGP+GlobeLand30+Knowledge) exhibits higher classification accuracy than FGP $2.0^{b}$ (FGP+DW+Knowledge). With the advancement of remote sensing land cover mapping, it is anticipated that more high-precision land cover products can be incorporated into the FGP 2.0 system, further enhancing the accuracy of the dual-driven mapping approach.

For the dynamic mapping task, FGP 2.0 optimizes the FGP Random Forest classification results using the updating annual map, classes of color-material, and DW dataset. While this mitigates classification errors and irrational land cover changes, some misclassifications persist. The limitations stem from the spectral characteristics-based classification methods of both the data-driven and knowledge-driven approaches used in dynamic mapping, as well as the integrated DW dataset. For instance, misclassifications related to water due to mountain shadows are partially alleviated but still present in certain areas. Exploring the incorporation of terrain information such as slope into the knowledge system to develop a knowledge graph enriched with advanced semantic details holds the potential to yield more rational classification results.

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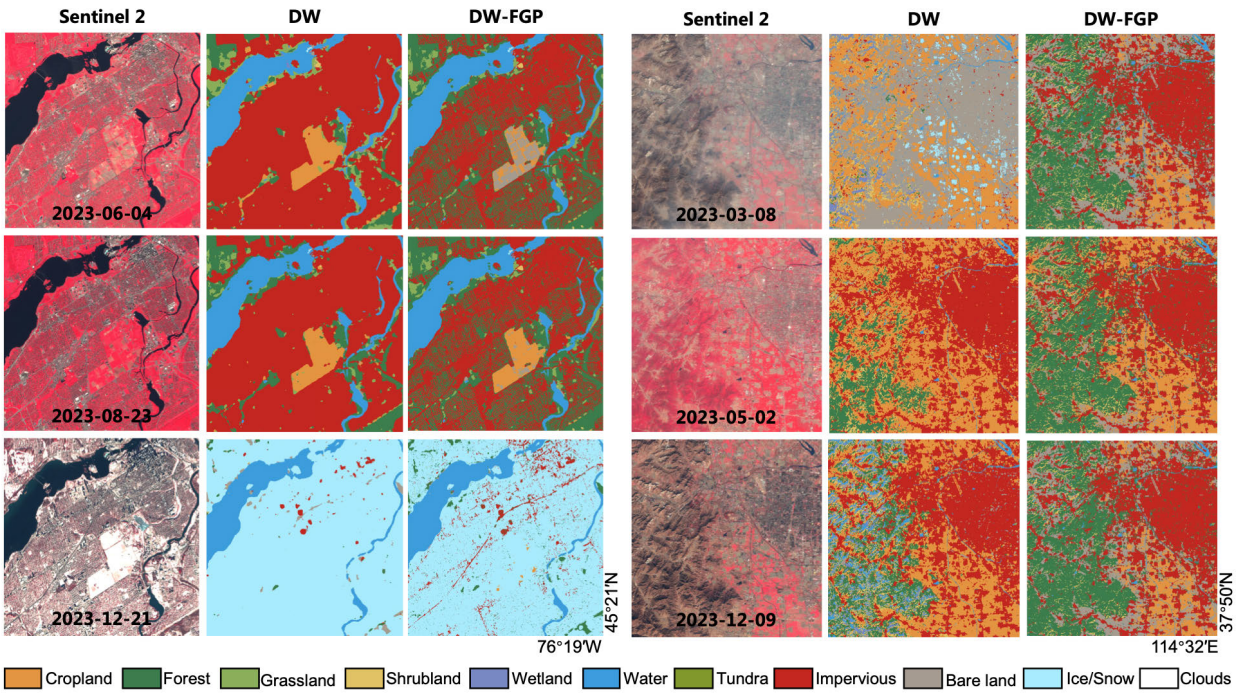



Fig. 9. Land cover map optimization results (DW-FGP) for DW using data and knowledge dual mapping framework (Sentinel-2 images are false-color rendered).

This knowledge-graph-based strategy can effectively organize geoscientific knowledge and provide more accurate reasoning and correction functions, which represents an ongoing focus of FGP 2.0.

In terms of data utilization, the current version of FGP 2.0 relies on Sentinel-2 imagery for both annual and dynamic mapping. Despite its high revisit frequency, regions prone to frequent cloud cover and rainfall, such as Southeast Asia, face limitations in acquiring adequate Sentinel-2 imagery [61], [62]. Therefore, integrating classification methods utilizing synthetic aperture radar (SAR) data, such as Sentinel-1, emerges as a crucial strategy to enhance the accuracy of land cover mapping in these challenging areas [63], [64], [65]. Additionally, SAR data contributes detailed texture features, proving effective in improving land cover classification [63], [64]. Incorporating SAR data into the FGP 2.0 mapping framework, for both of the Random Forest in FGP and SIAM in the ontology-based framework, represents a promising research direction with the potential to further elevate mapping precision.

## VI. Conclusion

In response to the accuracy challenges faced by current data-driven algorithms in land cover mapping products, this study introduces a knowledge and data dual-driven mapping framework (FGP 2.0), aiming to enhance both traditional annual mapping and emerging dynamic mapping. Building upon the foundation of the previous data-driven FGP, FGP 2.0 was developed as an intelligent remote sensing land cover mapping framework combined with knowledge-based reasoning and data learning. For annual mapping, FGP 2.0 significantly improves the overall accuracy of data-driven FGP (improved by ∼5% with a Z-score of 3.69), surpassing other land cover products such as DW and WorldCover (improved by about 16%–5% with Z-scores of 9.84 and 2.95). In dynamic mapping, FGP 2.0 offers more precise land cover distribution and reliable changes compared to FGP and DW (improved by about 20%–10% with Z-scores of 12.28 and 5.63). Moreover, with an extensible and flexible mapping framework, FGP 2.0 can be easily applied to optimize mapping products other than FGP. Therefore, with the integration of ontology-based semantic description and data-driven algorithms, FGP 2.0 proposed in this study bridges the gap between theoretical frameworks and practical applications. This intelligent land cover mapping framework meets the existing requirement for reliable land cover maps and offers an operational solution for merging geoscientific expertise with machine learning techniques.

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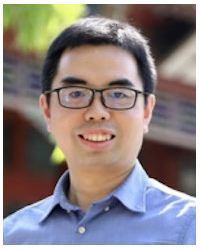

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